

Federated Learning Meets Knowledge Graphs:

Challenges and Opportunities for Distributed Health AI

1st International Workshop on Knowledge Graphs for Health

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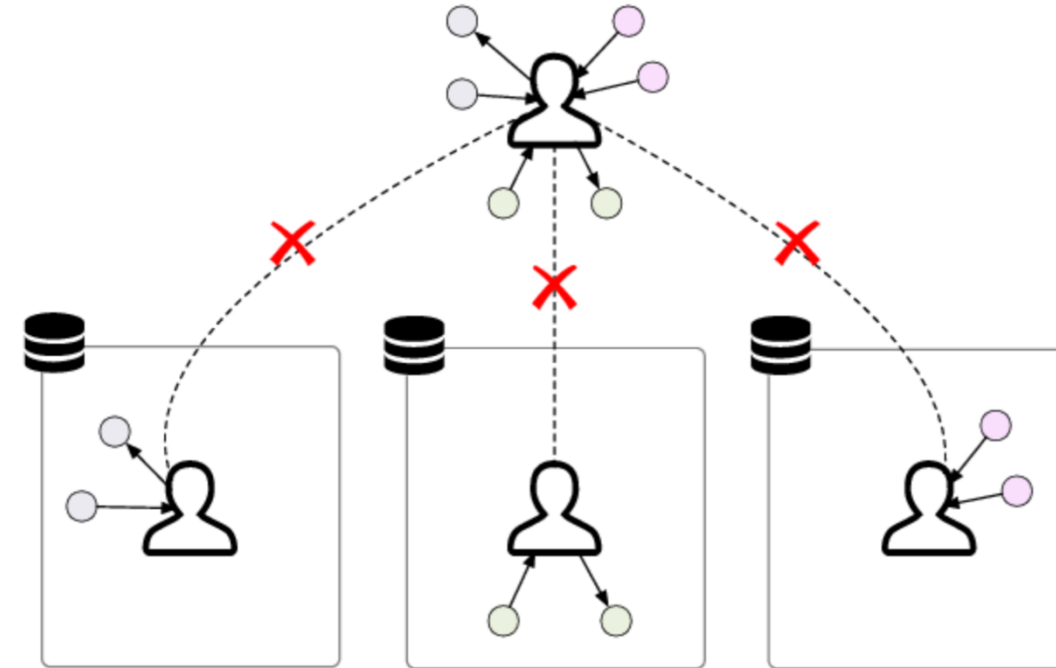
Motivation [1]

Problem

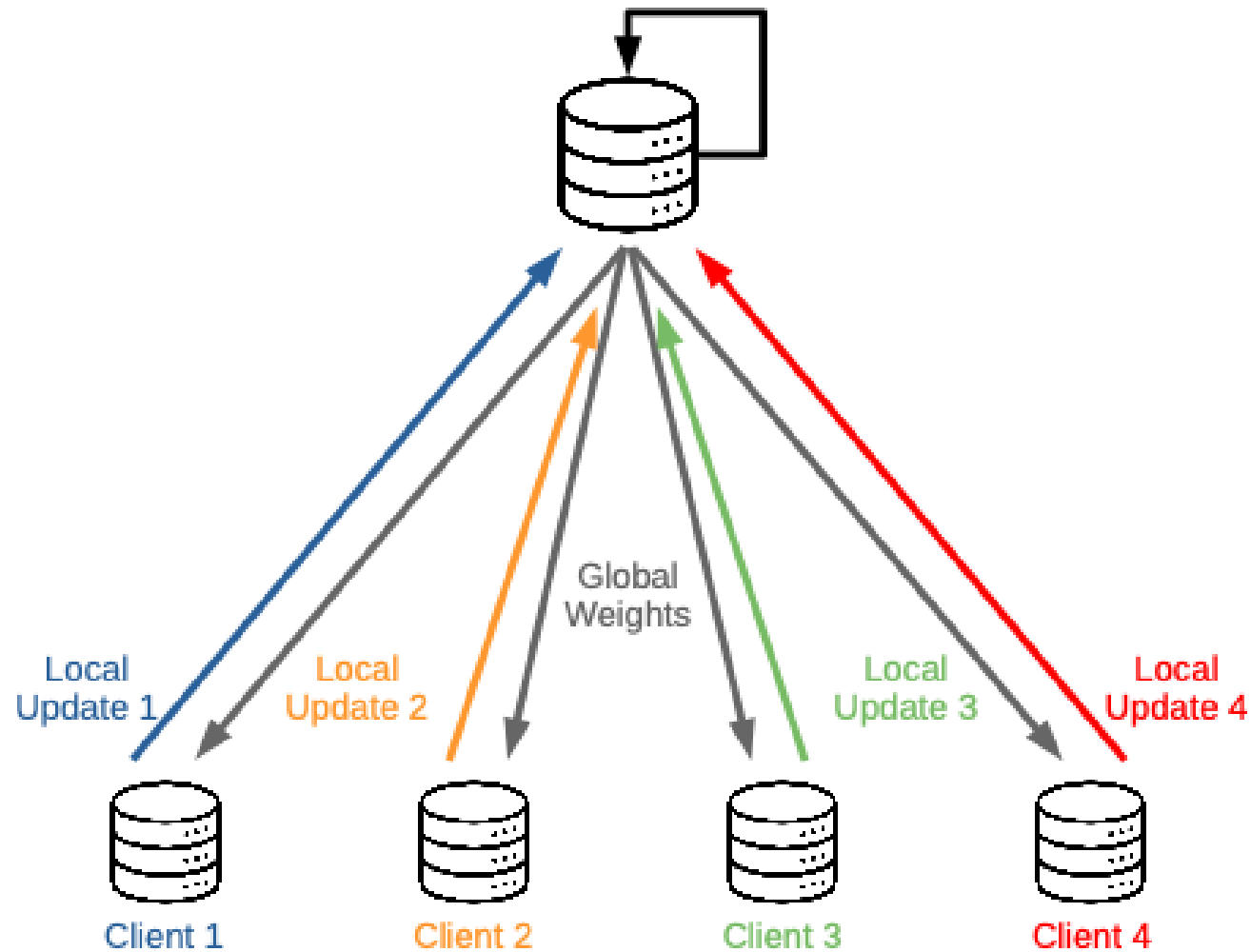
- ❖ Knowledge is spread across hospitals, biobanks, and public biomedical resources
- ❖ Each institution may hold only a partial view of the clinical knowledge space.
- ❖ Sensitivity of knowledge might make centralization unfeasible

Key question

- ❖ *“How can we learn from distributed health knowledge without requiring all data and graph structures to be centralized?”*



Federated Learning (FL)



From Federated Data to Federated Knowledge

Classical FL Setting	❖ Each client holds a local dataset ❖ Clients train a shared model on structurally similar data ❖ The server aggregates model updates
KG-based Setting	❖ Each client may hold a local graph ❖ Nodes, edges, relation types, and schemas may differ across clients ❖ The goal may be to learn embeddings, mappings, rules, or graph-based models without exchanging the full graph.
What changes?	❖ The distributed object is no longer only a dataset, but a knowledge representation ❖ Heterogeneity is not only statistical, but also semantic and structural. ❖ Alignment between entities, relations, and ontologies becomes part of the learning problem.

Opportunities

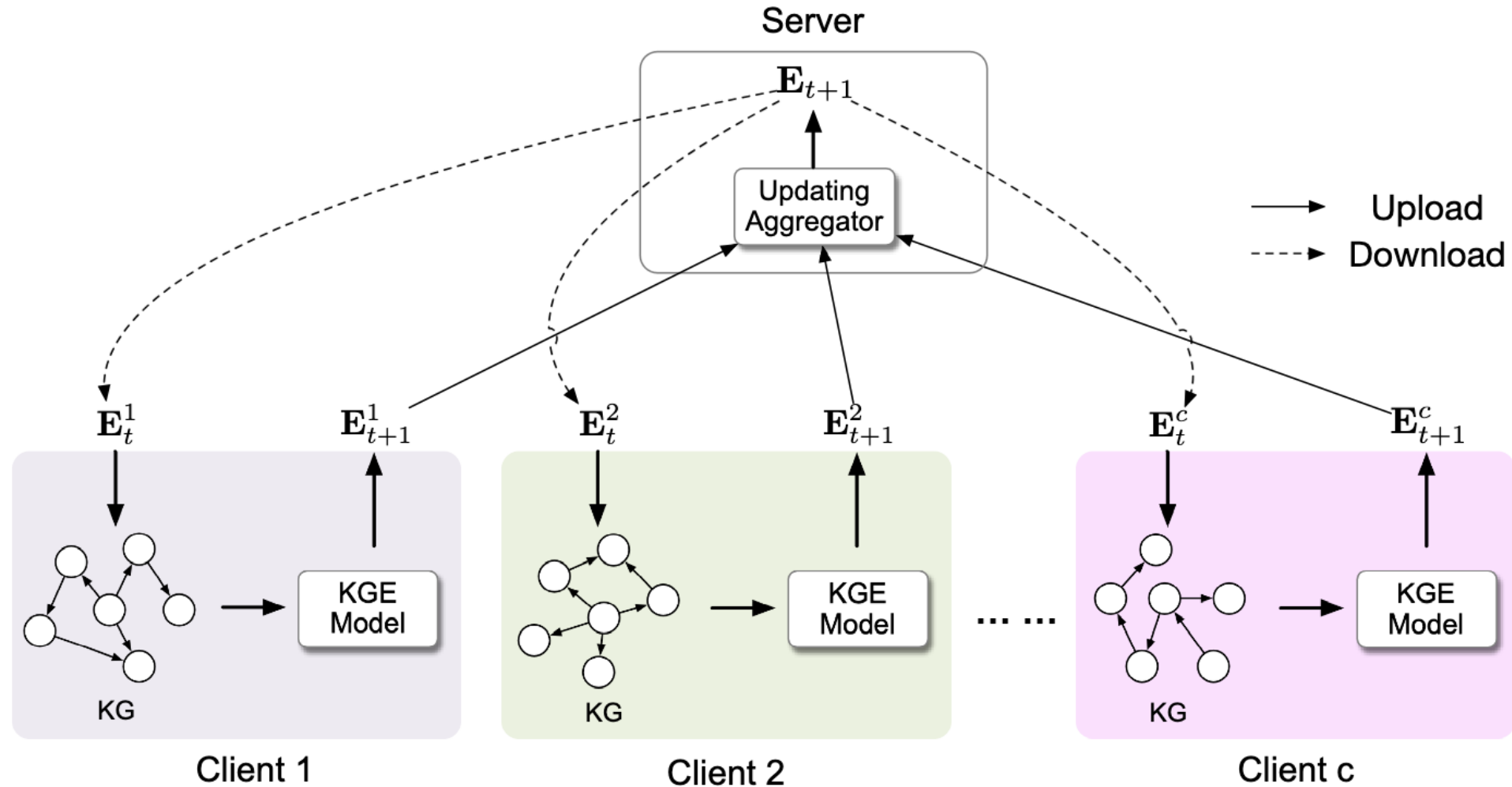
Opportunity 1:

Federated Knowledge Graph Embedding [1]

- Idea
- ❖ Instead of collecting all triples in one central KG, each institution keeps its local graph
 - ❖ A shared embedding model is trained collaboratively across clients.
 - ❖ Learn useful entity and relation representations without exchanging raw triples.
- Approach
- ❖ Local clients train a KG embedding model on their own triples.
 - ❖ Clients share model updates, embedding updates, or selected representation parameters.
 - ❖ The server aggregates updates to improve a global representation space.
- Applications
- ❖ Federated knowledge graph completion.
 - ❖ Link prediction across distributed graph fragments.
 - ❖ Learning from complementary institutional knowledge without central graph pooling.

Opportunity 1:

Federated Knowledge Graph Embedding [1]



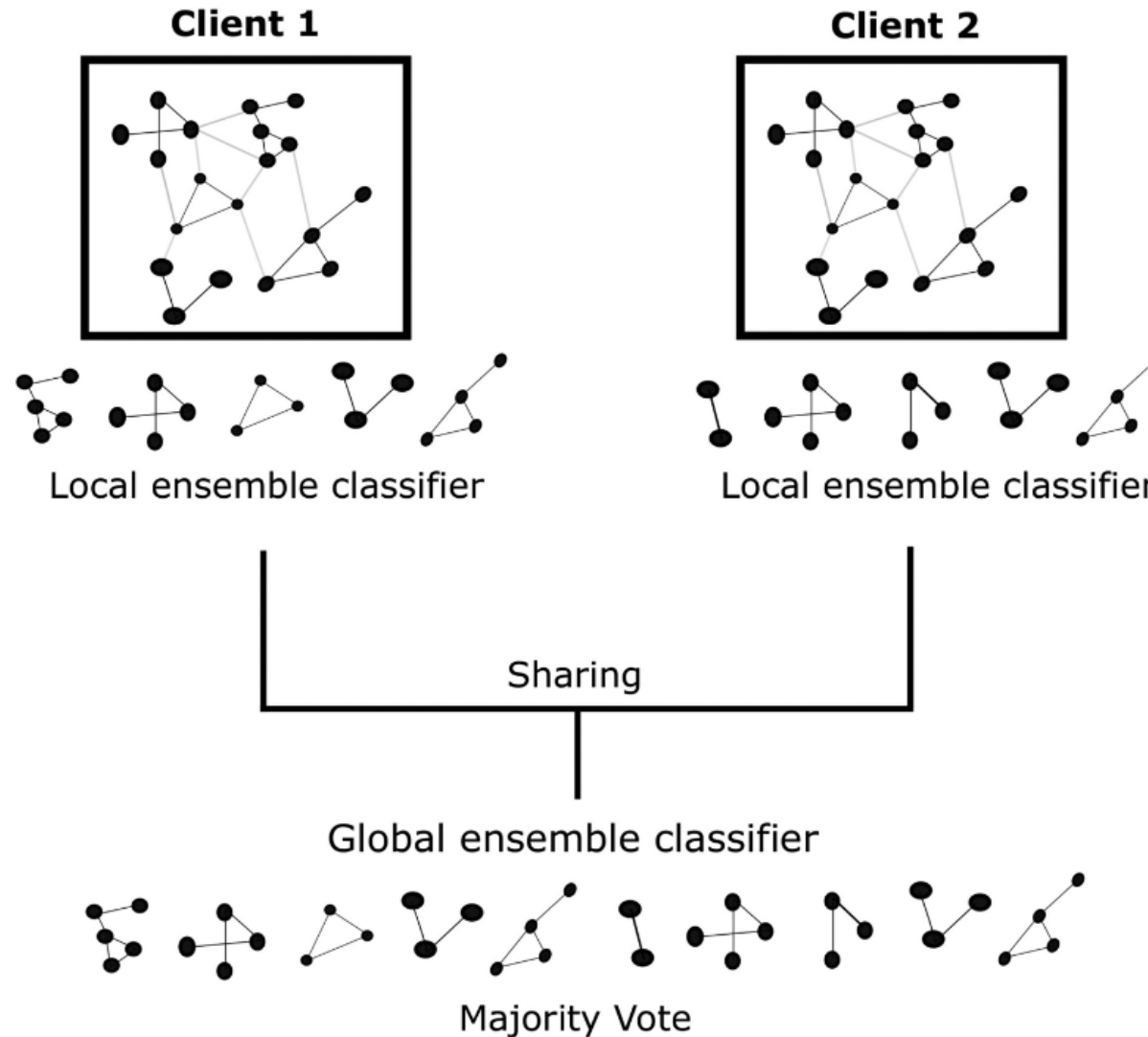
Opportunity 2:

Federated Graph Neural Networks (GNNs) [2]

- Idea
- ❖ Many biomedical problems are graph-structured
 - ❖ Instead of flattening these structures, GNNs can learn from nodes, edges and topology
 - ❖ FL allows multiple institutions to train graph-based models
- Approach
- ❖ Each client trains a local GNN (e.g., GCN) on graph structure
 - ❖ Much like in FL, clients only exchange model parameters (weights or gradients)
- Applications
- ❖ Disease classification from local omics and clinical data
 - ❖ Disease module discovery across institutions

Opportunity 2:

Federated Graph Neural Networks (GNNs) [2]



Opportunity 3:

Semantic Harmonization for Federated Learning [2]

Idea

- ❖ FL assumes that clients share the same feature space and label definitions.
- ❖ Sites may use different coding systems, terminologies, and measurement practices.
- ❖ KGs and ontologies can provide a shared semantic layer for aligning local data representations before or during FL.

Approach

- ❖ Each client maps local variables and concepts to a shared semantic reference model.
- ❖ The shared layer may include ontologies, concept hierarchies, and mappings.
- ❖ Clients may exchange semantic mappings, feature correspondences, ontology-aligned metadata, or model updates over harmonized features.

Opportunity 3:

Semantic Harmonization for Federated Learning [2]

Example

- ❖ Site A: “HbA1c”
- ❖ Site B: “Glycated hemoglobin”
- ❖ Site C: “LOINC 4548-4”
- ❖ => *All referring to the same clinical concept, but different specificity*

Solution

- ❖ A shared KG/ontology layer can map these to a common concept while preserving local data ownership.

Challenges

Challenge 1:

Statistical Heterogeneity Becomes Graph Heterogeneity

- ❖ In classical FL, data imbalance is a major concern
- ❖ In KG settings, heterogeneity is even richer:
 - ❖ Different entities
 - ❖ Different relations
 - ❖ Different schemas
 - ❖ Different graph densities
 - ❖ Different ontology depth
 - ❖ Different missingness patterns
- ❖ => *In federated KGs, clients may not just have different data distributions; they may represent the world differently*

Challenge 2:

What is Actually Shared?

- ❖ In FL, the exchanged object is often clear: model parameters or gradients
- ❖ In KG settings, this is less obvious. Possible exchange objects are:
 - ❖ GNN parameters
 - ❖ Node embeddings
 - ❖ Ontology mappings
 - ❖ Graph summaries
 - ❖ Constraints or rules
- ❖ => *What is the “model update” when the underlying object is a knowledge graph?*

Challenge 3:

Privacy Leakage from Graph Structure

- ❖ Even if raw patient data is not shared, graph representations may leak information
- ❖ Potential leakage sources
 - ❖ Rare entities
 - ❖ Unique neighborhoods
 - ❖ Sensitive edges
 - ❖ Embeddings that encode identifiable relations
 - ❖ Membership inference
 - ❖ Reconstruction attacks
- ❖ => *Keeping the graph local does not automatically make the system private*

Conclusion

Main message

- ❖ FL and KGs address complementary parts of the same problem:
how to build useful health AI when data and knowledge is distributed?

Opportunities

- ❖ Learn shared representations from distributed KGs
- ❖ Train graph-based prediction models
- ❖ Use KGs and ontologies as semantic infrastructure for cross-site harmonization

Challenges

- ❖ Clients may differ not only statistically, but also structurally and semantically
- ❖ It is unclear what should be exchanged: parameters, embeddings, mappings, ...
- ❖ Keeping graphs local does not guarantee data privacy

Thank you for your attention!



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- [1] Chen, M., Zhang, W., Yuan, Z., Jia, Y., & Chen, H. (2021, December). Fede: Embedding knowledge graphs in federated setting. In *Proceedings of the 10th international joint conference on knowledge graphs* (pp. 80-88).
- [2] Pfeifer, B., Chereda, H., Martin, R., Saranti, A., Clemens, S., Hauschild, A. C., ... & Heider, D. (2023). Ensemble-GNN: federated ensemble learning with graph neural networks for disease module discovery and classification. *Bioinformatics*, 39(11), btad703.
- [3] Kokash, N., Wang, L., Gillespie, T. H., Belloum, A. S., Grosso, P., Quinney, S., ... & De Bono, B. (2026). Ontology-and LLM-based data harmonization for federated learning in healthcare. *Frontiers in Digital Health*, 8, 1756555.